shows the patient's name and address, as well as the procedures that were performed, and the total charge for the procedures.

Assume you are writing an application to generate a statement that can be printed and given to the patient.

A) Identify all of the potential classes in this problem domain.
B) Refine the list to include only the necessary class or classes for this problem.
C) Identify the responsibilities of the class or classes that you identified in step B.

Review Questions and Exercises

Short Answer

1. What is the difference between a class and an instance of the class?
2. What is the difference between the following `Person` structure and `Person` class?

```
struct Person
{
   string name;
   int age;
};
```

```
class Person
{
   string name;
   int age;
};
```

3. What are the attributes and methods of an object?
4. Look at the following constructor definition:

   ```
   Date :: Date ( int dd = 1, int mm = 1, int yy = 0  );
   ```

   What is the name of the class?

   What type of a constructor is this?
5. A contractor uses a blueprint to build a set of identical houses. Are classes analogous to the blueprint or the houses?
6. What is a mutator function? What is an accessor function?
7. Is it a good idea to make member variables private? Why or why not?
8. Can you think of a good reason to avoid writing statements in a class member function that use `cout` or `cin`?
9. Under what circumstances should a member function be private?
10. What is a constructor? What is a destructor?
11. What is a default constructor? Is it possible to have more than one default constructor?
12. Is it possible to have more than one constructor? Is it possible to have more than one destructor?
13. If a class object is dynamically allocated in memory, does its constructor execute? If so, when?

14. What does the following statement in an UML class diagram mean?

    `+ getVal(num : int) : void`

15. What are a class's responsibilities?
16. How do you identify the classes in a problem domain description?

Fill-in-the-Blank

17. The two common programming methods in practice today are _________ and _________.
18. Combining data and methods into a single object is called _________.
19. _________ is an object's ability to hide its data from code that is outside the object.
20. _________ is an object's ability to contain and manipulate its own data.
21. In C++, the _________ is the construct primarily used to create objects.
22. _________ specify how class members may be accessed.
23. An object's _________ defines the values that are stored in the object's attributes at any given moment.
24. The default access specification of class members is _________.
25. Classes and _________ have the same relationship as data types and variables.
26. Defining a class object is often called the _________ of a class.
27. In the statement `double Rectangle::getArea()`, `Rectangle` is the name of the _________ and '`::`' is called the _________.
28. If you were writing the declaration of a class named `Canine`, what would you name the file it was stored in? _________
29. If you were writing the external definitions of the `Canine` class's member functions, you would save them in a file named _________.
30. When a member function's body is written inside a class declaration, the function is _________.
31. A(n) _________ is automatically called when an object is created.
32. A(n) _________ is a member function with the same name as the class.
33. _________ are useful for performing initialization or setup routines in a class object.
34. Constructors cannot have a(n) _________ type.
35. In a class without constructors, a _________ is written during compilation.
36. A(n) _________ is a member function that is automatically called when an object is destroyed.
37. A destructor has the same name as the class, but is preceded by a(n) _________ character.
38. Like constructors, destructors cannot have a(n) _________ type.
39. A constructor whose arguments all have default values is a(n) _________ constructor.
40. Defining more than one constructor in a class is called _________.
41. A class may only have one default _________ and one _________.
42. _________ provides a set of standard diagrams for graphically representing object-oriented systems.

Algorithm Workbench

43. Write a class declaration named `Circle` with a private member variable named `radius`. Write set and get functions to access the `radius` variable, and a function named `getArea` that returns the area of the circle. The area is calculated as

    `3.14159 * radius * radius`

44. Add a default constructor to the `Circle` class in Question 43. The constructor should initialize the `radius` member to 0.
45. Add an overloaded constructor to the `Circle` class in Question 44. The constructor should accept an argument and assign its value to the `radius` member variable.
46. Write a statement that defines an array of five objects of the `Circle` class in Question 45. Let the default constructor execute for each element of the array.
47. Write a statement that defines an array of five objects of the `Circle` class in Question 45. Pass the following arguments to the elements' constructor: 12, 7, 9, 14, and 8.
48. Write a `for` loop that displays the radius and area of the circles represented by the array you defined in Question 47.
49. If the items on the following list appeared in a problem domain description, which would be potential classes?

Animal	Medication	Nurse
Inoculate	Operate	Advertise
Doctor	Invoice	Measure
Patient	Client	Customer

50. Look at the following description of a problem domain:

    The bank offers the following types of accounts to its customers: savings accounts, checking accounts, and money market accounts. Customers are allowed to deposit money into an account (thereby increasing its balance), withdraw money from an account (thereby decreasing its balance), and earn interest on the account. Each account has an interest rate.

    Assume you are writing an application that will calculate the amount of interest earned for a bank account.

    A) Identify the potential classes in this problem domain.
    B) Refine the list to include only the necessary class or classes for this problem.
    C) Identify the responsibilities of the class or classes.

True or False

51. T F Private members must be declared before public members.
52. T F Class members are private by default.
53. T F Members of a `struct` are private by default.
54. T F Classes and structures in C++ are very similar.
55. T F It is mandatory to declare all data members in a class to be private.
56. T F All public members of a class must be declared together.
57. T F It is legal to define a pointer to a class object.
58. T F A `const` member function cannot modify any data members.

59. T F Private data members cannot be accessed from outside a class unless one knows the password to access it.
60. T F Constructors do not have to have the same name as the class.
61. T F Constructors are automatically executed when an object is created.
62. T F The compiler will report an error if a `cin` statement is in its body.
63. T F Destructors cannot take arguments.
64. T F Constructors and destructors are executed inline.
65. T F Constructors may have default arguments.
66. T F Member functions may be overloaded.
67. T F Constructors may not be overloaded.
68. T F Constructors and destructors have the same syntax and can only be differentiated from each other through the code that is written in its body.
69. T F A class may only have one destructor.
70. T F When an array of objects is defined, the constructor is only called for the first element.
71. T F If an inline function is called, and the compiler finds that inline expansion is not possible, the compiler ignores the request.
72. T F In an UML diagram, public data members are indicated by a ‘ – ’ and private data members are indicated by a ‘ + ’.

Find the Errors

Each of the following class declarations or programs contain errors. Find as many as possible.

73.
```
class Triangle()
{
private
   double side1, side2, side3;
public
   getsides();
   setsides(s1, s2, s2);
   double findarea();
   void display();
}
```

74.
```
#include <iostream>
using namespace std;

Class Moon;
{
Private;
   double earthWeight;
   double moonWeight;
Public;
   moonWeight(double ew);
      { earthWeight = ew; moonWeight = earthWeight / 6; }
   double getMoonWeight();
      { return moonWeight; }
}
```

```
int main()
{
   double earth;
   cout >> "What is your weight? ";
   cin << earth;
   Moon lunar(earth);
   cout << "On the moon you would weigh "
        <<lunar.getMoonWeight() << endl;
   return 0;
}
```

75.
```
#include <iostream>
using namespace std;

class DumbBell;
{
   int weight;
public:
   void setWeight(int);
};
void setWeight(int w)
{
   weight = w;
}
int main()
{
   DumbBell bar;

   DumbBell(200);
   cout << "The weight is " << bar.weight << endl;
   return 0;
}
```

76.
```
class List
{
   int number[20];
   int loop;
   Date()
   {  loop = 0; }
   List(int n) ;
};
List :: List (int n)
  {
     loop = n;
     cout << "Initialize numbers in list ---" << endl;
     for ( int i = 0; i<loop ; i++)
     {
        cout << "Enter number : ";
        cin >> number[i];
     }
}
```